CMUC Online Seminar on Algebra, Logic and Topology

University of Coimbra

Algebraically cocomplete categories

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Abstract

Dualizing the concept introduced by Freyd in 1990, a category is called algebraically cocomplete if every endofunctor has a terminal coalgebra. Among cocomplete categories these are just preordered classes. However, assuming the generalized continuum hypothesis (GCH), we present nice examples: the category Set_k of sets of power at most k is algebraically cocomplete for all uncountable regular cardinals k. (In contrast, the category of countable sets is algebraically complete, but not cocomplete.) Also the category Vec_k of vector spaces of dimension a most k is algebraically cocomplete and, given a finite group G, so is the category G-Set_k of G-sets of power at most k. The assumption of GCH cannot be simply omitted: we prove that the continuum hypothesis is *equivalent* to the statement that the category of sets of power at most aleph_1 is algebraically cocomplete.